

Algebra 1 - Lab 1: Tiles → Algebra

Name: _____ Period: _____

Module 1 — Like terms and the Distributive Property

Materials: algebra tiles · pencil. Every problem:

BUILD it → SORT it → CANCEL zero pairs → COUNT it.

$\boxed{x}$ = x $\boxed{1}$ = 1 $\boxed{-x}$ = -x $\boxed{-1}$ = -1 shaded = flipped (negative)

Part 1 — Build it, sort it, cancel, count. A minus means a FLIPPED tile. Sort bars with bars, squares with squares. A tile and its flipped twin are a ZERO PAIR — cancel them. Count what is left.

a. $3x + 2 + 2x + 5$ workspace — build here bars = _____ squares = _____ simplified: _____ ★ Think deeper: A bar and a square are BOTH tiles. Why can't you count them together as one number? _____ _____	b. $6y + 4 - 2y - 1$ workspace — build here bars = _____ squares = _____ simplified: _____ ★ Think deeper: What if there were MORE flipped bars than plain bars? What is left after cancelling, and what does it mean? _____ _____	c. $8 + 3n - 5 + n$ workspace — build here bars = _____ squares = _____ simplified: _____ ★ Think deeper: Write a DIFFERENT jumbled expression that builds the exact same two piles. _____
---	---	--

Part 2 — Rows. The number in front of the parentheses tells you how many ROWS to build. Every row is a copy of the inside.

a. $3(2x + 1)$ workspace — build here bars = _____ squares = _____ simplified: _____ ★ Think deeper: Build $3 \cdot 2x + 1$ — same numbers, NO parentheses. Is it the same picture? Why not? _____ _____	b. $2(4t + 3) - 5t$ workspace — build here bars = _____ squares = _____ simplified: _____ ★ Think deeper: Could you cancel the zero pairs BEFORE building all the rows? Would you get the same answer? _____ _____	c. $4(x + 2) + 3x$ workspace — build here bars = _____ squares = _____ simplified: _____ ★ Think deeper: Compare the picture for $4(x + 2) + 3x$ with the picture for $3x + 4(x + 2)$. Same piles? Name the property. _____ _____
--	---	--

Part 3 — Flip the row. A minus sign in front of a row flips EVERY tile in it. Then cancel zero pairs exactly as before.

a. $-(2y + 3)$ workspace — build here bars = _____ squares = _____ simplified: _____ ★ Think deeper: Flip the flipped row again. What expression is that, and what does it tell you about $-(-a)$? _____ _____	b. $-2(3n - 4)$ workspace — build here bars = _____ squares = _____ simplified: _____ ★ Think deeper: Why does the -4 inside become $+8$? Explain with tiles, not a rule. _____ _____	c. $10k - 3(2k - 1)$ workspace — build here bars = _____ squares = _____ simplified: _____ ★ Think deeper: What would the picture look like if the answer had NO bars at all? Write such an expression. _____ _____
--	--	--

Part 4 — Team Challenge, then the No-Tiles Check. Work with your group on a-c. Then put the tiles AWAY for the check on the right.

a. Ana simplifies $3(2x + 4)$ and gets $6x + 12$. Ben gets $6x + 4$. Build it. Who is right, and what did the other person forget? _____ _____ ★ Think deeper: Is there ANY number outside the parentheses where Ben's shortcut still gives the right answer? _____ _____	b. Write the expression this picture shows TWO ways: with parentheses, and simplified. <table border="1"> <tbody> <tr> <td>x</td><td>x</td><td>x</td><td>1</td><td>1</td></tr> <tr> <td>x</td><td>x</td><td>x</td><td>1</td><td>1</td></tr> </tbody> </table> _____ c. Design one: write an expression with parentheses AND a subtraction whose finished picture has 9 bars and 6 squares. Trade with a partner and build theirs. _____ _____	x	x	x	1	1	x	x	x	1	1	No-Tiles Check. Pencil only. See the tiles in your head; write the simplified expression. 1. $5(2n + 3) - 4n$ = _____ 2. $9k - 3(2k - 7)$ = _____ 3. $7m + 4 - 3m - 9$ = _____ 4. $-4(x - 5) + 6x$ = _____ ★ Think deeper: Which of the four takes the MOST tiles to build? Is it also the hardest? Why or why not? _____ _____
x	x	x	1	1								
x	x	x	1	1								